Project Specifications

The group was assigned the task of building an automatic subway turnstile using a 555 Timer, a J-K Flip-Flop and any necessary logic gates (AND, OR, XOR, NAND, NOR and NOT gates). Furthermore, the automatic turnstile must accept a ticket in its' slot to unlock the gate and allow a customer through. If the customer uses the turnstile or waits over 10 seconds after entering a ticket the latch must return to its' locked state. The system must be able to continue to lock and unlock depending on the customers actions without failure [1]. The basic block diagram of the timing system can be seen below in Figure 1.

Figure 1. Basic System Block Diagram [1].

Note that the system should be designed to minimize complexity by using the minimum amount of logic gates required to implement the system. For a full description of the assigned project see Appendix A.

Design Process and Implementation.

To simulate the automatic turnstile the group chose to use 2 buttons representing the customers actions, Button 1 representing entering a ticket and Button 2 representing the customer using the turnstile. As well, the group chose to use two LEDs to represent the Free or Locked state of the turnstile, where LED 1 (Green) represents the Free state and LED 2 (Red) represents the Locked state.

To get a basic understanding of the 555 Timer the group designed the desired system without the J-K Flip Flop as can be seen below in Figure 2.

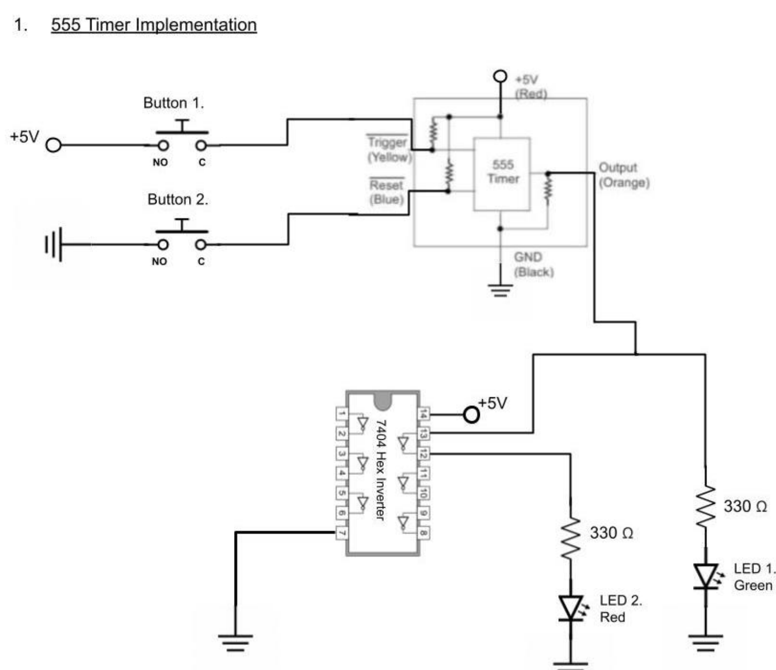


Figure 2. System Implementation with just the 555 Timer [2] [1].

The group wired Button 1 from a 5V supply to the Trigger input and Button 2 from Ground to the Reset input on the 555 Timer. Note that the 7404 Hex Inverter is used to invert the output from the 555 Timer To ensure that LED 1 and 2 are never simultaneously on. When Button 1 is pressed and released the 555 Timer's output will go from Low to High, turning LED 2 off and LED 1 on, until the Reset button is hit, or the timer reaches around 10 seconds. After the Timer resets the Output will go back to Low turning off LED 1 and turning on LED 2. The physical circuit can be seen below in Figure 3.

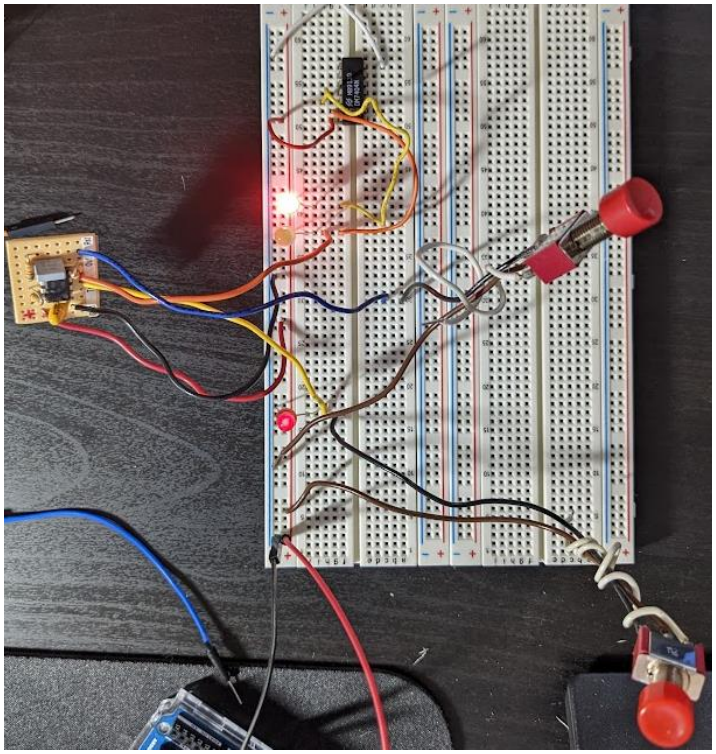
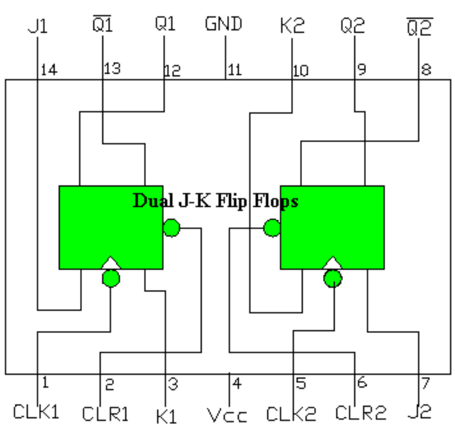


Figure 3. 555 Timer Physical Circuit Implementation.

Although this configuration seems to meet the system requirements, the inserting action should not be dependent on the release of the button, since in a real system the customer either enters the ticket ot doesn't. Further the group can use a J-K Flip-Flop to send a signal pulse from the users button press rather than waiting for the customer to release the button. To implement a J-K Flip-Flop the group used the 7473 Dual Master-Slave Chip as can be seen below in Figure 4 with its truth table.



Function Table

Inputs				Outputs	
CLR	CLK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	⎍	L	L	Q_0	$\overline{Q}_0$
H	⎍	H	L	H	L
H	⎍	L	H	L	H
H	⎍	H	H	Toggle	

H = HIGH Logic Level
L = LOW Logic Level
X = Either LOW or HIGH Logic Level
⎍ = Positive pulse data. the J and K inputs must be held constant while the clock is HIGH. Data is transferred to the outputs on the falling edge of the clock pulse.
Q_0 = The output logic level before the indicated input conditions were established.
Toggle = Each output changes to the complement of its previous level on each HIGH level clock pulse.

Figure 4. Dual Master-Slave J-K Flip-Flop Pin Out Diagram and Truth Table [3].

To produce a pulse from pressing Button 1 our group decided to utilize the Clear Function on the Flip-Flop. As can be seen by the Function Table by setting J to High and K to Low if Clear is High the output after the falling edge of the Clock Pulse is Q = High, $\bar{Q}$ = Low. As well, if Clear is set to Low the output regardless of the other inputs is Q =Low, $\bar{Q}$ = High. Therefore, the group can use Q to output the pulse.

However, to utilize the Clock function on J-K Flip-Flop the system must input some type of square wave into the Clock Pin. Therefore, the group decided to utilize propagation delay by wiring 3 NOT gates (using the 7404 Hex Inverter) from Button 1 to the Clock input. Note that these inverters are in parallel with the Button's input to the Clear function. This allows the J-K Flip-Flop to function off of the delay between the Clear and Clock inputs to output a pulse each time Button 1 is pressed. To see a hand draw circuit diagram, see Appendix B. A timing diagram of the system including the propagation delay can be seen below in Figure 5.

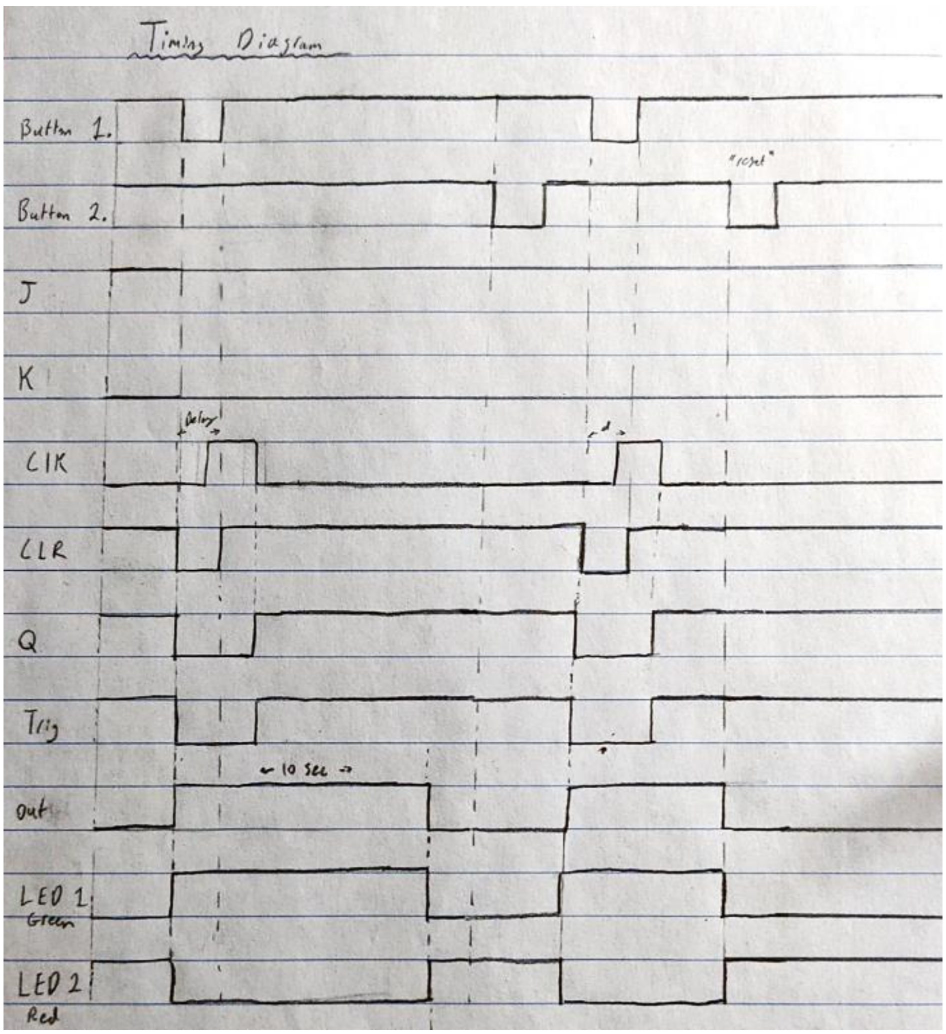


Figure 5. System Timing Diagram after Button Presses.

As shown above there is a delay in the pulse of the Clock and Clear from the series of NOT gates that allows the J-K Flip Flop to utilize the falling clock edges properly. Note that the Buttons and Clear inputs are in an active Low state.

Final Circuit Diagram

The final design of the Turnstile can be seen below in Figure 6. Note that the NOT gates are implemented using the 7404 Hex Inverter chip and the J-K Flip Flop was implemented using the 7473 Dual Master-Slave Chip. As well, note that the group chose to use 330 Ohm resistors paired with the LEDs to reduce the impact of any voltage surges that may cause the LEDs to break.

2. JK Flip Flop and 555 Timer Implementation

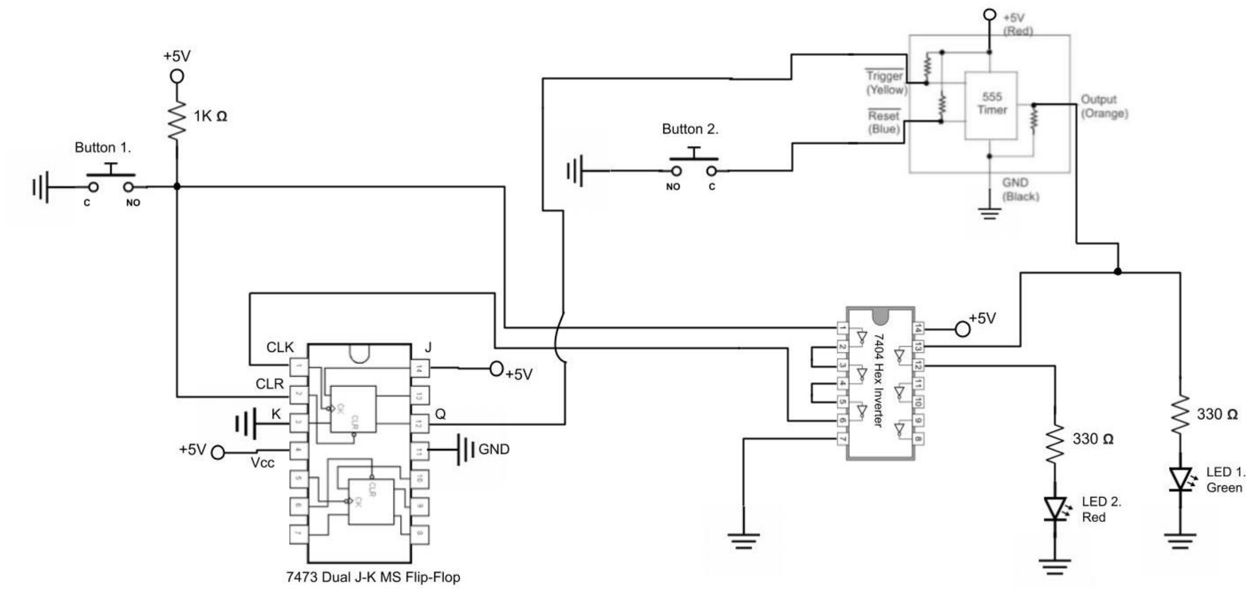


Figure 6. Final System Circuit Diagram [4] [1] [2].

To view a simplified circuit diagram without the Hex Inverter displayed see Figure 7 below.

3. Simplified JK Flip Flop and 555 Timer Implementation

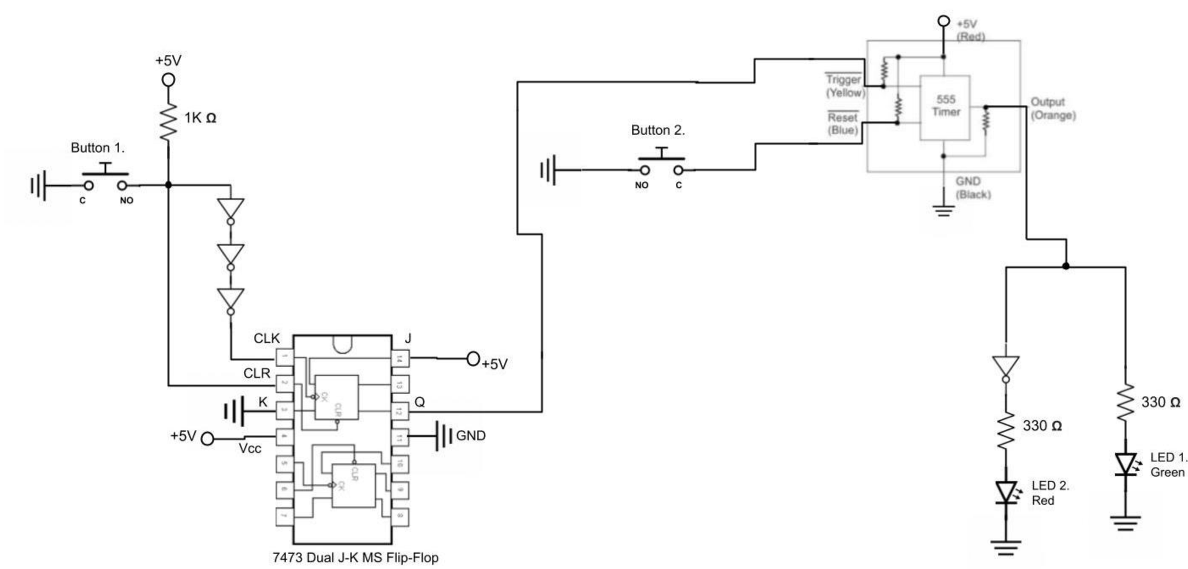


Figure 7. Simplified Circuit Diagram [1] [4].

The physical circuit can be seen below in Figure 8, where the 7404 Hex Inverter chip is on the first breadboard and the 7473 J-K M/S Flip-Flop is on the 2nd breadboard.

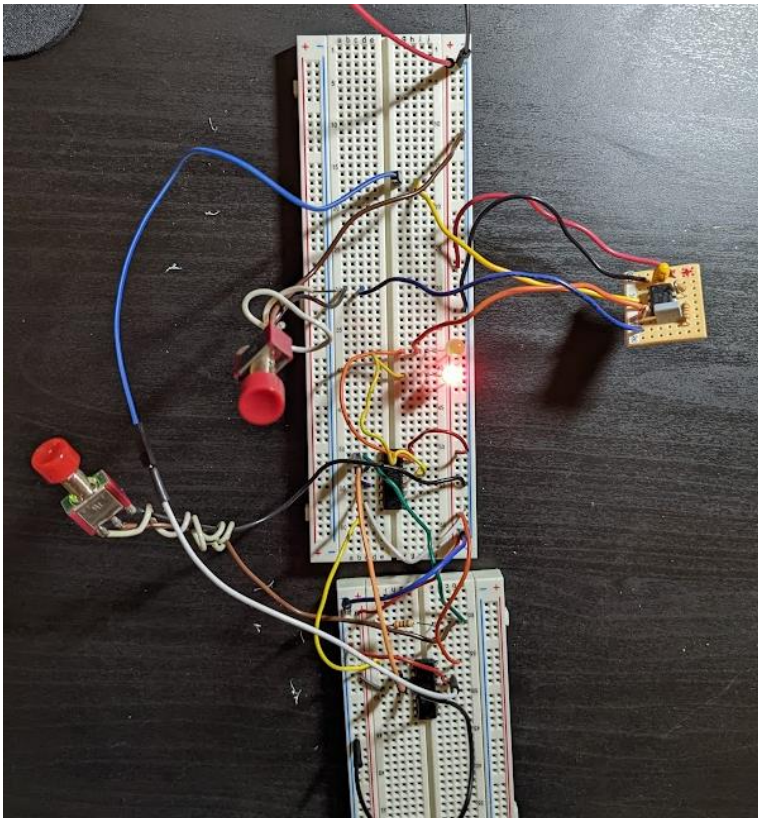


Figure 8. Physical Final System Design.

References

[1] B. Shastri, "ENPH 334 Specifications for Digital Project," Queen's University, Kingston, 2022.

[2] Wiki Media Commons, "File:7404 Hex Inverters.PNG," Wiki Media Commons, 14 April 2009. [Online]. Available: https://commons.wikimedia.org/wiki/File:7404_Hex_Inverters.PNG. [Accessed 20 November 2022].

[3] Make Your Own Chip, "7473 DATASHEET Dual Master-Slave J-K Flip-Flops," Make Your Own Chip, [Online]. Available: https://makeyourownchip.tripod.com/7473.html. [Accessed 19 November 2022].

[4] SRM University, " EC1010 Digital Systems Lab_Manual," SRM University, [Online]. Available: https://webstor.srmist.edu.in/web_assets/srm_mainsite/files/downloads/EC1010_digital_systems.pdf. [Accessed 19 Novewmber 2022].

Appendix A. Given Design Description.

Problems need to include a latch specifically the SN7473 J-K flip-flop, either to inhibit the input signal for a time, or to send out a signal to operate some other mechanism. Failure to include one will reduce the total amount of marks.
In these problems, you must deal with just a small part of the overall system. Do not bring in other factors which would just add unnecessary complexity.

D. An automatic exit turnstile at a subway station accepts the ticket in a slot, and a latch produces a high signal and enables the turnstile for a predetermined time. If the turnstile is not operated during this time, a high signal is sent to lock the turnstile and return the ticket so that the customer may try again. Not that the enable and lock systems are different controls requiring separate high signals. Design the system to meet these specifications.

Appendix B. Hand Drawn System Diagrams

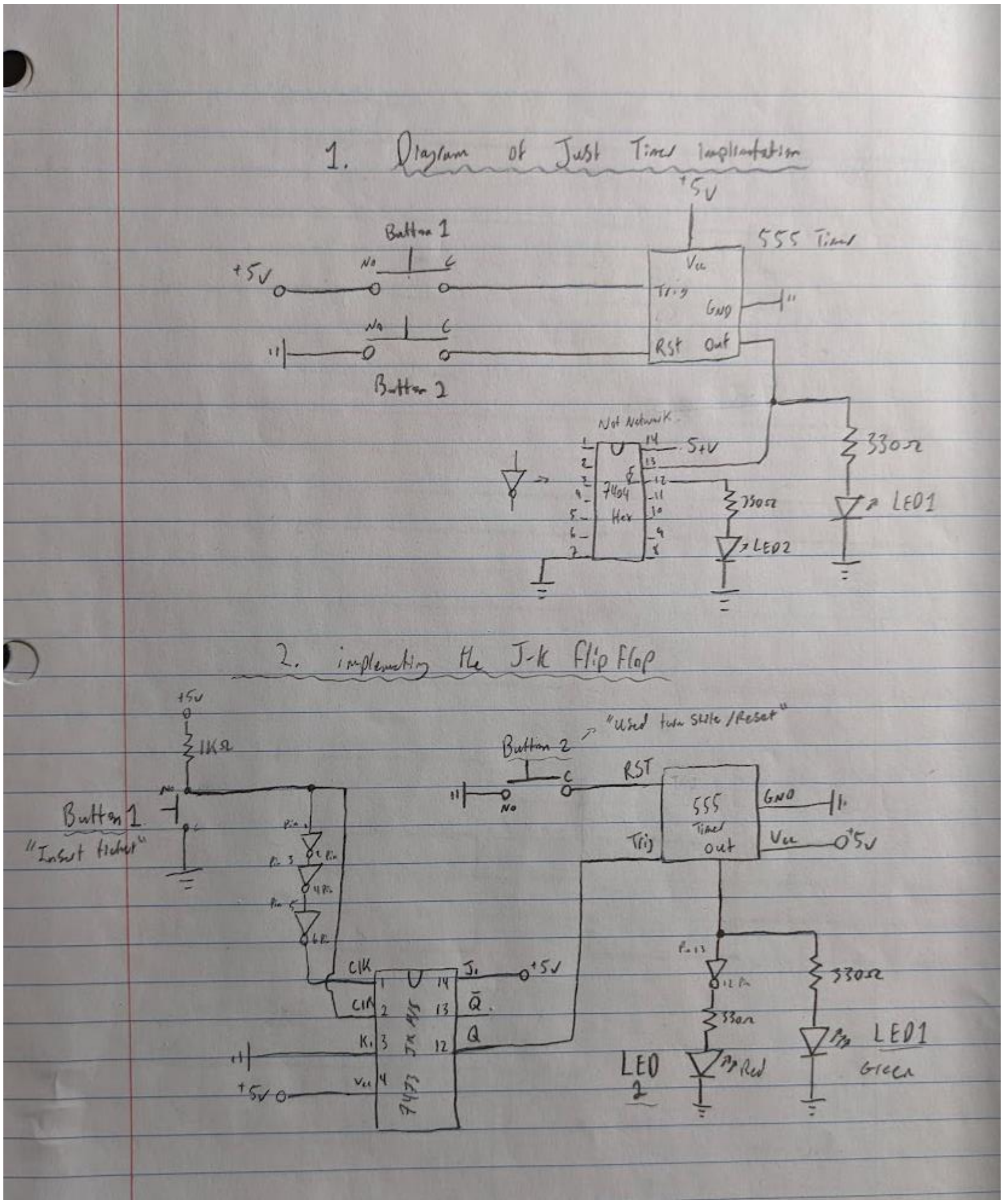


Figure 9. Hand drawn circuit diagram from lab notebook.